

1. Administrative & Study Identification

Full Study Title: A Phase 1/2 Study to Evaluate the Safety and Efficacy of Patritumab Deruxtecan in Gastrointestinal Cancers
Short Title/Acronym: HERTHENA-PanTumor02 (MK-1022-011) **Protocol Number:** MK-1022-011 **NCT Number:** NCT06596694 **Trial Status:** Recruiting **Sponsor/Company:** Merck (Merck Sharp & Dohme LLC in collaboration with Daiichi Sankyo) **Investigational Product:** Patritumab deruxtecan (MK-1022 / HER3-DXd / U3-1402) **Phase of Development:** Phase 1/Phase 2 **Medical Monitor:** Manik Amin, MD (Lead Physician) / Merck Clinical Trials

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of patritumab deruxtecan and to evaluate the confirmed objective response rate (ORR) per RECIST v1.1 assessed by blinded independent central review (BICR) for each cohort.
Secondary Objectives: To evaluate the duration of response (DOR) and progression-free survival (PFS) per RECIST v1.1 by BICR and overall survival (OS), and to characterize the pharmacokinetics of patritumab deruxtecan.

2.2 Background & Rationale A significant medical need exists for more effective and tolerable therapeutic options for patients with advanced gastrointestinal cancers, specifically colorectal cancer (CRC), biliary tract cancer (BTC), and hepatocellular carcinoma (HCC). Human epidermal growth factor receptor (HER3) expression is high in CRC and BTC, and low-to-intermediate in HCC. Patritumab deruxtecan is a HER3-directed antibody-drug conjugate (ADC) incorporating a topoisomerase I inhibitor payload. The study aims to examine its antitumor activity, safety, and efficacy in these select gastrointestinal cancers.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm / Single Group Assignment **Blinding:** Open-label (Masking: None) **Randomization:** Nonrandomized (Allocation: N/A)

3.2 Planned Enrollment & Duration Targeted Enrollment: 180 adult participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 03-Nov-2024 **Estimated Primary Completion:** 07-Dec-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Has one of the following previously treated cancers: Unresectable or metastatic colorectal adenocarcinoma, advanced and/or unresectable biliary tract cancer (BTC), hepatocellular carcinoma (HCC) not amenable to locoregional therapy, or locally advanced unresectable or metastatic gastroesophageal cancer. 3. Has received prior therapy for the cancer. 4. Has recovered from any side effects due to previous cancer treatment.

4.2 Exclusion Criteria 1. Has a history of (noninfectious) interstitial lung disease (ILD)/pneumonitis that required steroids, has current ILD/pneumonitis, or suspected ILD/pneumonitis cannot be ruled out by standard diagnostic assessments at screening. 2. Has clinically severe respiratory compromise (based on the investigator's assessment) resulting from intercurrent pulmonary illnesses. 3. Has evidence of any leptomeningeal disease. 4. Has clinically significant corneal disease. 5. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease, or evidence of ongoing uncontrolled systemic bacterial, fungal, or viral infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patritumab deruxtecan administered as monotherapy. Cohorts 1 and 2 will receive standard dosing; Cohort 3 (HCC) includes a dose-escalation phase to determine the recommended phase 2 dose. **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Continues until documented disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptoms such as nausea and pain. **Prohibited:** Prior topoisomerase I inhibitor therapies are strictly prohibited. Concomitant use of other investigational systemic anticancer therapies is restricted.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Pulmonary Evaluation
Baseline	Day 0	Cohort Allocation, First Dose (IV Infusion), Safety Labs
Interim Visits	Q3W (Generic ADC)	IV Infusions, Safety Labs, Efficacy Assessment (Tumor Scans), PK draws
End of Study	Follow-up	Final Vital Signs, Exit Interview, Off-treatment safety follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of Participants Experiencing Dose-Limiting Toxicity (DLT):** Evaluated during the Dose-Escalation Phase. Defined as any drug-related adverse event resulting in a dose change or delay of the next cycle.
- 2. Number of Participants with One or More Adverse Events (AEs):** Defined as any unfavorable and unintended sign, symptom, disease, or worsening of preexisting condition temporally associated with study treatment.
- 3. Number of Participants who Discontinue Study Intervention Due to an AE.**

7.2 Secondary & Exploratory Endpoints

- 1. Duration of Response (DOR)** assessed by BICR.
- 2. Progression-Free Survival (PFS)** defined as the time from the first day of study intervention to the first documented progressive disease (PD) or death, assessed by BICR.
- 3. Overall Survival (OS).**
- 4. Pharmacokinetics (PK)** profile of patritumab deruxtecan.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events, with stringent protocols for the early detection and management of Interstitial Lung Disease (ILD) / pneumonitis.

Serious Adverse Events (SAEs): Reported within 24 hours per standard regulatory oncology procedures.

Data Monitoring Committee (DMC): Utilized to monitor dose-escalation (Cohort 3) and oversee global safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy assessed in the Intention-to-Treat (ITT) population; safety analyzed in the Safety Set (all patients receiving ≥ 1 dose).

Sample Size Justification: $N = 180$ allows for the detection of preliminary efficacy and establishing safety profiles within individual histologically defined cohorts (CRC, BTC, HCC, gastroesophageal).

Handling of Missing Data: Standard censoring methods applied for time-to-event endpoints (PFS, OS) based on RECIST v1.1 conventions.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular clinical monitoring including source data verification (SDV) focusing on RECIST measurements and critical safety events like ILD.

Audit Trail: Electronic Case Report Form (eCRF) tracking with complete audit trails and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-1022-011 **Posting ID:** 240547923952254660 **Registry Number:** NCT06596694 **Sponsor/Lead Organization:** Merck **Study Title:** Study of Patritumab Deruxtecan in Participants With Gastrointestinal Cancers (MK-1022-011) (HERTHENA-PanTumor02) **Official Regulatory Title:** A Phase 1/2 Study to Evaluate the Safety and Efficacy of Patritumab Deruxtecan in Gastrointestinal Cancers

12. Clinical Framework (Oncology Specific)

Disease Area / Name: Gastrointestinal Cancer **Preferred Name:** Malignant neoplasm of gastrointestinal tract **Preferred UMLS Name:** Gastrointestinal Neoplasms **Semantic Type:** Neoplastic Process **Definition:** A primary or metastatic malignant neoplasm involving any part of the digestive system. **Diagnosis Threshold:** Advanced, unresectable, or metastatic status **Medical Methodology:** HER3-directed antibody-drug conjugate (ADC) targeting via topoisomerase I inhibitor payload **Ontology Codes:** MeSH: D005770 | HPO: HP:0006749 | SNOMED: 428905002

13. Study Procedures & Methodology

Management Pathway: Standard oncological pathways for Phase 1/2 advanced solid tumors. **Education Protocol (HCP):** Site training on prompt identification and management of ADC-related toxicities (e.g., ILD/pneumonitis protocol). **Education Protocol (Patient):** Symptom tracking for respiratory compromise and reporting guidelines for potential treatment toxicities. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all tumor imaging to assure unbiased response rate calculations.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4 years (Estimated primary completion December 2028) **Onsite Visit Cadence:** Concurrently mapped to treatment cycles (typically IV infusions) **Remote Monitoring Frequency:** Continuous pharmacovigilance tracking **Checklist Review Interval:** Scheduled tumor assessment intervals (e.g., every 6-8 weeks) per RECIST v1.1.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	Manik Amin, MD	____	[DD-MMM-YYYY]				

| | Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | |
Lead Statistician | [Name] | _____ | [DD-MMM-YYYY] |